

Haocheng He

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PROFESSIONAL SUMMARY

IE undergrad at GDUT, now doing an MEng in aviation at PolyU. Along the way I did 4 internships — a Shenzhen startup where I worked on smart piano hardware and AI chips, an airport simulation project with Guangzhou Baiyun that won a thesis award, and two factory stints at Midea and LiuGong doing time studies and line balancing. Filed a few patents along the way. I like figuring out how things actually work and making them run better.

EDUCATION

The Hong Kong Polytechnic University Hong Kong SAR, China
Master of Engineering in Aviation Engineering and Operation Management
Sep 2025 – Jan 2027
GPA: 3.95 / 4.3 (Current)

Guangdong University of Technology Guangzhou, China
Bachelor of Engineering in Industrial Engineering
Sep 2021 – Jun 2025
GPA: 3.67 (Top 20%) — Excellent Graduation Thesis + Innovation Award

Minor in Law Sep 2022 – Jun 2025
GPA: 3.61

EXPERIENCE

AI Operations & R&D Intern May 2025 – Apr 2026
Shenzhen Wuxian Hongyin Technology Co., Ltd. Shenzhen, China

- Deployed AI inference workflows on Huawei Ascend 310B chips for internal team use; improved response time by approximately 30%.
- Designed motor driver circuits (LCEDA) and wrote STM32 firmware for a smart piano; tested with oscilloscopes and logic analyzers.
- Developed a finger-position tracking system using camera input and the Ascend chip; authored SDK technical documentation.
- Oversaw server operations (Ubuntu), network infrastructure (Huawei switches/routers), procurement processes, and technical interviews.

Research Assistant — Airport Operations Simulation Sep 2023 – Jun 2025
Guangdong University of Technology (with Guangzhou Baiyun Int'l Airport) Guangzhou, China

- Modeled ground vehicle operations at Guangzhou Baiyun Airport using AnyLogic, Java, and MySQL, covering road networks, service lanes, and dispatch rules.
- Implemented routing algorithms (Dijkstra, A*, BFS, genetic) for 50+ vehicles across 3 zones to find collision-free paths.
- Reduced simulated congestion by 35% during peak hours (200+ movements/hour) with zero collisions.
- Thesis awarded Excellent Graduation Thesis and Innovation Award by Guangdong University of Technology.

Industrial Engineering Intern Apr 2024 – May 2024
Midea Group — Kitchen & Water Heater Division Foshan, China

- Conducted stopwatch time studies across all workstations on a dishwasher sheet metal line; identified 5 steps exceeding cycle time.
- Proposed 22 improvements using lean methods; simulation showed line balance improving from 58% to 70%, reducing headcount from 20 to 18.

Manufacturing Operations Intern Jul 2023 – Aug 2023

- Timed 12 workstations on the wheel loader assembly line with a stopwatch; identified 3 major bottlenecks using ECRS analysis.
- Proposed layout adjustments estimated to reduce cycle time by approximately 8% in simulation. Discussed findings with line engineers.

SKILLS

Programming: Python, Java, SQL, STM32 (C)
Tools & Platforms: AnyLogic, Plant Simulation, OR-Tools, MySQL, LCEDA, oscilloscope, logic analyzer
Languages: English (IELTS 6.5), Mandarin (native), Cantonese (conversational)

PUBLICATIONS & INTELLECTUAL PROPERTY

Summary: 3 invention patents + 1 utility model + 2 software copyrights from research at GDUT.

Invention Patents

- PI Coating Machine Motion Control via MCMC Sampling — Zhou Xianzhong, He Haocheng, He Zhiyuan (2026). CN122085794A, Guangdong University of Technology.
- Coating Control Method and System (Deep Reinforcement Learning) — Zhou Xianzhong, He Haocheng, He Zhiyuan (2025). CN121325552A, Guangdong University of Technology.
- Home Solar Energy Storage System — Adaptive Energy Regulation — Zhou Xianzhong, Zeng Huangtao, Yu Daming, He Haocheng (2024). CN117767437A, Guangdong University of Technology.

Utility Model

- Integrated Wire Harness Production Equipment — Zhao Xiang, He Haocheng, Hu Bofeng (2024). CN221352455U, Guangdong University of Technology.

Software Copyright

- Enterprise Financial Data Analysis Software V1.0 — Li Yisi, He Haocheng (2024). Registration No. 2024SR0146827.
- Urban Flood Intelligent Mobile Emergency Command Dispatch System V1.0 — Lin Chengfeng, Li Zeyin, Yang Hua, He Haocheng (2024). Registration No. 2024SR0549247.

Awards

- Excellent Bachelor Thesis and Innovation Award — Haocheng He (2025). Guangdong University of Technology.